

MATH 1342 – Elementary Statistics – Week 2: Sampling, Experiments and Observational Studies

Aug 31 – Sep 2 | 1.2 Sampling from a Population • 1.3 Experiments and Observational Studies | Fall 2026 · Section 019 · MW 7:30–8:50 PM · RRC Room 8210

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Distinguish a population from a sample and describe statistical inference. (CLO 1) 2. Identify sampling bias in a described study, classify the sampling method (random, convenience, volunteer), and judge whether the results can be generalized. (CLO 1) 3. Recognize bias from question wording, question context, and inaccurate responses. (CLO 1)	e-Text: 1.2 Sampling from a Population Interactive Lecture: 1.2 Sampling from a Population Slides: 1.2 Sampling from a Population (includes sampling video)	Participation: Variables in Statistics (Kahoot!) – Mon: review of Week 1 Participation: Sampling – spot the bias in sample scenarios Participation: Sampling Methods (Kahoot!) – Wed: sampling review	Online Homework: 1.2 Sampling from a Population (due Sun Sep 6, 11:59 PM)	WileyPLUS StatKey Lecture Slides (course website) Kahoot! In class
4. Classify a study as observational or experimental, and identify a plausible confounding variable. (CLO 1) 5. Explain how random assignment, a control group, and blinding support cause-and-effect conclusions, and distinguish random sampling (generalize) from random assignment (causation). (CLO 1)	e-Text: 1.3 Experiments and Observational Studies Interactive Lecture: 1.3 Experiments and Observational Studies Slides: 1.3 Experiments and Observational Studies (includes TED-Ed placebo video)	Participation: Experiments and Observational Studies – classify studies; confounding; random sampling vs. random assignment	Online Homework: 1.3 Experiments and Observational Studies (due Sun Sep 6, 11:59 PM)	WileyPLUS StatKey Lecture Slides (course website) In class
<i>Items that cover objectives from more than one row above (objectives in parentheses)</i>			Take-Home Quiz 1 (due Wed Sep 9, 11:59 PM) (all objectives)	Assignment Tool in Blackboard

CLO 1 Explain the use of data collection and statistics as tools to reach reasonable conclusions.